

Control of *Listeria monocytogenes* ATCC 19115 biofilm formation and virulence by a food-derived *Lactobacillus acidophilus* VBFDP771 isolate with predictive assessment of robustness

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ABSTRACT

Biofilm formation by foodborne pathogens such as *Listeria monocytogenes* significantly contributes to their persistence in food-related environments and resistance to conventional control strategies. The use of food-derived lactic acid bacteria (LAB) represents a promising biological approach for mitigating pathogenic biofilms and associated virulence traits. In this study, a *Lactobacillus acidophilus* VBFDP771 isolate obtained from traditional yogurt was evaluated for its ability to control *L. monocytogenes* ATCC 19115 biofilm formation and suppress virulence-associated gene expression, while assessing its physiological robustness under gastrointestinal-relevant conditions. The isolate exhibited strong tolerance to acidic conditions (pH 3–5) and bile salts (0.2–0.6%), maintaining a viable count of 7.45 log CFU/mL after simulated gastrointestinal transit. It showed notable surface hydrophobicity (60.85%), auto-aggregation (43.61%), co-aggregation (46.63%), and adhesion to Caco-2 intestinal cells (12.63%), supporting its functional stability. The strain significantly inhibited *L. monocytogenes* adhesion through competitive exclusion (46.71%), inhibition (40.88%), and displacement (26.86%) mechanisms. Treatment with cell-free supernatant markedly reduced biofilm biomass and resulted in substantial downregulation of key virulence- and biofilm-associated genes (*sigB*, *flaA*, *inlB*, *agrA*, *hly*, *prfA*, and *plcB*). In addition, the isolate demonstrated antioxidant activity (DPPH: 62.85%; ABTS: 63.55%) and cholesterol assimilation capacity (58.80%), and exhibited a favorable safety profile, including the absence of hemolytic activity, DNase production, and biogenic amine synthesis, as well as acceptable antibiotic susceptibility. Gaussian Process Regression (GPR) modeling further supported strain robustness by accurately predicting acid and bile tolerance responses ($R^2 = 0.97$ and 0.94 , respectively). Overall, these findings highlight a food-oriented biological strategy for controlling *L. monocytogenes* persistence through biofilm inhibition and virulence attenuation, while demonstrating the utility of integrating functional and predictive assessments for food safety applications.

1. Introduction

The term probiotic was first introduced in the early 20th century, derived from a phrase meaning for life. Probiotics are live microorganisms that, when consumed in adequate amounts—typically between 10^6 and 10^7 colony forming unit (CFU)/mL can significantly impact human health (Alizadeh Behbahani, Jooyandeh & Namazi, 2024; Alizadeh Behbahani, Jooyandeh, Hojjati and Ghodsi Sheikhjan, 2024; Alizadeh

Behbahani, Noshad, Namazi and Vasiee, 2024). They promote well-being by balancing intestinal flora and fostering interactions between the gastrointestinal microbiota and the immune system. Probiotics can secrete antimicrobial substances, adhere to the mucous membranes and epithelial cells, and enhance the intestinal epithelial barrier (Zibaei-Rad et al., 2023; Topçu, Kaya and Kaban, 2020). Key benefits of consuming probiotic bacteria include alleviating lactose intolerance, reducing inflammation and the risk of cancers affecting the

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intestines and liver, lowering blood cholesterol levels, boosting the immune system, preventing infections, treating and preventing diarrhea and constipation, enhancing the absorption of vitamins and minerals, and inhibiting the growth of harmful bacteria (Vasiee et al., 2020; Falah et al., 2021).

Most probiotics are derived from lactic acid bacteria (LAB), with *Lactobacillus acidophilus* being one of the most notable. This gram-positive bacterium is characterized by its spherical and rod shapes, facultative anaerobic nature, catalase negativity, and inability to produce spores or indole, as well as being negative for cytochrome oxidase and nitrate reduction. Like other LAB, *L. acidophilus* can ferment carbohydrates to produce lactic acid, diacetyl, and bacteriocins (Noshad et al., 2021). It is primarily a fermentative bacterium, with lactic acid as its main metabolic product, making it well-suited for use in probiotic dairy products like yogurt. The *L. acidophilus* bacterium exhibits robust proliferation when maintained at a pH range of 5.5–5.8, coupled with temperatures spanning 30–40 °C, and an atmosphere containing 5% carbon dioxide (Alizadeh Behbahani et al., 2022). To ensure optimal efficacy, the strain must endure the challenging conditions of the human gastrointestinal system. Consequently, multiple laboratory analyses are crucial for fully characterizing its beneficial probiotic attributes. These examinations gauge the bacterium's resistance to high acidity and bile salts, its viability within the gut environment, its capacity for attachment to host intestinal epithelia, its ability to diminish cholesterol levels, its role in breaking down bile salts, its antagonistic actions against pathogenic and non-hemolytic microbes, its confirmed absence of DNase enzyme synthesis, and its stability during processing and storage (Vasiee et al., 2020).

Listeria monocytogenes is the bacterium responsible for listeriosis, which can occur after consuming contaminated foods, particularly red meat. It is notable for its ability to thrive at low temperatures, even within refrigerator ranges (1–45 °C), and across a wide pH spectrum (4.3–9.8) (Echresh et al., 2024; Alizadeh Behbahani, Noshad, Vasiee and Brück, 2024). Can also form biofilms, which are complex microbial communities comprised of various microorganisms that adhere to surfaces and enhance survival (Moradi et al., 2020). These biofilms are notably resistant to antimicrobial agents and can contribute to infectious diseases in humans. To eliminate *Listeria monocytogenes* and its biofilms, several preservation strategies are employed, including heat treatment, chemical disinfectants, mechanical removal, and the use of safe, natural antimicrobial compounds. Notably, employing LAB as a preservation method is both environmentally friendly and effective in inhibiting the growth or destroying pathogenic microorganisms (Falah et al., 2024; Rouhi et al., 2024).

Modeling techniques have become indispensable in food science and technology, offering powerful tools for optimizing processes, ensuring quality, and enhancing safety. Among these techniques, Artificial Neural Networks (ANNs) have gained significant attention due to their ability to model complex, non-linear relationships without requiring explicit mathematical formulations (Alizadeh Behbahani, Noshad and Vasiee, 2024). ANNs mimic the human brain's neural structure, enabling them to learn from data and make accurate predictions based on patterns and trends. In food science, ANNs are applied across various domains, including the prediction of microbial growth, sensory evaluation, and the optimization of food processing parameters. For instance, ANNs can predict the shelf life of products, assess food quality through image analysis, and optimize fermentation processes (Daliran et al., 2023). Their flexibility and robustness make them ideal for handling the diverse and dynamic nature of food systems. The integration of ANNs in food technology not only improves efficiency but also contributes to sustainability by reducing waste and energy consumption. As the food industry continues to evolve, the role of ANNs in driving innovation and ensuring food safety and quality is expected to expand further (Majdi et al., 2024). Some researchers applied ANN to predict some food characterizes and reported that ANN can predict the output with high accuracy (Rahmati-Joneidabad et al., 2025; Zibaei-Rad et al., 2024;

Faradonbeh et al., 2024; Majdi et al., 2024).

The aim of this study was to investigate the probiotic properties of the *L. acidophilus* strain VBFDP771. The investigation tested its endurance against stomach and gut conditions, including tolerance to acidity and bile salts, and assessed its capacity for cholesterol reduction and antioxidative characteristics. Antimicrobial performance was quantified using multiple established techniques, such as disc diffusion and minimum inhibitory concentration tests. Additionally, the study analyzed anti-adhesion strategies (competition, inhibition, and displacement), aggregation potential, and affinity for Caco-2 cells. Safety was confirmed by analyzing cytotoxicity, DNase function, non-hemolytic status, and biogenic amine generation, alongside its antibiotic resistance phenotype. Additionally, the Gaussian Process Regression (GPR) algorithm was implemented to forecast concentrations of both acidity and bile salts.

2. Materials and methods

2.1. Chemicals

The chemicals and reagents utilized in this study were of analytical grade and purchased from Sigma Co. (USA) or Merck Co. (Germany).

2.2. Probiotic properties evaluation methods

2.2.1. Isolation and identification

Briefly, yogurt samples (Behbahan, Iran) were mixed with peptone water for homogenization before being cultured on de Man-Rogosa-Sharpe (MRS) agar. Following culture, strain extraction involved Gram staining, catalase tests, and genomic DNA extraction. The 16S rRNA gene was amplified using specific universal primers, with PCR conditions optimized for successful segment amplification, which was then confirmed by agarose gel electrophoresis. The analysis revealed the isolate as *L. acidophilus* VBFDP771, showing 99% similarity based on nucleotide sequencing (Access code: ACCESSION MW479186) (Kivanc et al., 2011).

2.2.2. Tolerance to pH

The tolerance under varying pH conditions was assessed using a suspension of LAB (10^9 CFU/mL) in phosphate buffered saline (PBS), which was adjusted to a pH range of 3–5 through the addition of HCl, following the method of Balcázar et al. (2008). This was done for 0–3 h hours at 22 °C. After the pH treatment, samples were serially diluted in sterile PBS and incubated, and viable counts were then determined through plate counting using MRS (Balcázar et al., 2008).

2.2.3. Simulated gastrointestinal tract stability

The viability of the *L. acidophilus* VBFDP771 strain was investigated and evaluated using an in vitro model simulating human gastrointestinal (GIT) conditions. The LAB strain was cultured for 18 h, subsequently washed twice with PBS at pH 7.4, and then resuspended to a standardized concentration of OD600 at 1. A 1 mL aliquot of this final suspension was combined with 10 mL of simulated gastric juice (GJ) containing 3 g/L pepsin (pH 3.0). This mixture was incubated for 2 h at 37 °C with shaking at 100 rpm. Cells treated in GJ were then harvested by centrifugation and resuspended in simulated intestinal juice, which contained 0.3% bile salt and 0.1% pancreatin. This secondary treatment proceeded for 2 h at 37 °C under anaerobic conditions. Finally, viable cell counts were enumerated before the start of each phase and upon the completion of the entire GIT transit (Huang et al., 2015).

2.2.4. Tolerance to bile salts

The bacterial culture was centrifuged and the pellets were washed with phosphate-saline buffer, then resuspended in sterile MRS broth containing varying concentrations of bile salt (0.2, 0.4 and 0.6% w/v). Cell survivability was assessed by collecting samples at 0-, 1-, 2-, and 3-

hours post-incubation, alongside a positive control in plain MRS broth without bile salts. After incubation, cultures were diluted and plated on MRS agar, which was incubated at 37 °C under anaerobic conditions for 24 to 48 h. The results were reported as colony-forming units per milliliter (CFU/mL) (Mulaw et al., 2019).

2.2.5. Hydrophobicity

Bacterial adherence to hydrocarbons was evaluated using a modified method from Somashekaraiah et al. (2019). *L. acidophilus* cells were cultured in MRS broth at 37 °C for 18 h, then centrifuged and washed with phosphate urea magnesium (PUM) buffer. The cell pellet was resuspended in PUM buffer and adjusted to an optical density of approximately 0.7 at 600 nm before being mixed with n-hexadecane. After phase separation, the absorbance of the aqueous phase was measured at 600 nm to calculate the surface hydrophobicity percentage.

$$\text{Hydrophobicity}\% = [(A_{\text{initial}} - A_{\text{final}}) / A_{\text{initial}}] \times 100$$

2.2.6. Auto-aggregation

The culture from overnight was gathered using centrifugation (at 8000 rpm for 10 min at 4 °C), subsequently rinsed twice with PBS, and then re-suspended in PBS buffer. The sample was permitted to settle and was incubated at 37 °C. After 5 h, the absorbance at 600 nm of the upper suspension was recorded. The calculation of auto-aggregation percentage was performed using the following formula:

$$\text{Auto - aggregation}\% = [1 - (A_5 / A_0) \times 100]$$

where A₅ signifies the absorbance at 5 h and A₀ refers to the absorbance at time 0 (Somashekaraiah et al., 2019).

2.2.7. Co-aggregation

The co-aggregation assay was performed using a spectrophotometer (WPA Biowave II, UK) and based on the protocol of Keller et al. (2011). First, the concentration of the strains was standardized by adjusting the OD₆₀₀ to 0.5 (approximately equivalent to ~10⁸ cells/mL). Then, equivalent volumes of the probiotic and control suspensions were mixed and incubated at a temperature of 37 °C. Finally, the co-aggregation ratio was determined and calculated after 5 h of incubation, by measuring the optical density of the pathogen suspension in comparison with the mixed suspension.

2.2.8. Anti-infection properties

To evaluate the probiotic potential of the *L. acidophilus* VBFD771 strain in inhibiting pathogen attachment, its anti-adhesion inhibitory effects against the bacterium *Listeria monocytogenes* on the Caco-2 intestinal cell monolayer were assessed. To this end, in vitro anti-infective assays including Co-incubation, Pre-incubation Inhibition, and Post-infection Displacement were performed (Bendali et al., 2014).

2.2.9. Adhesion

The adhesion capabilities of the *L. acidophilus* VBFD771 strain using the Caco-2 cell line were performed in accordance with the methods of Kim et al. (2015). Initially, the Caco-2 cell line was cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% FBS. Then, for the evaluation of adhesion, the cell monolayers were incubated with the bacterial suspension (2 × 10⁸ CFU/mL) in DMEM medium without phenol red for 2 h at 37 °C in a 5% CO₂ atmosphere. After washing with PBS, the Caco-2 cells were lysed with 1% cold Triton X-100 to release the adhered bacteria. Finally, the number of released bacteria was determined by performing serial dilutions and plating them onto MRS agar medium.

2.2.10. Antimicrobial activity

The antibacterial properties of *L. acidophilus* VBFD771 were assessed using various methodologies, including well diffusion agar,

disk diffusion agar, modified double-layer technique, and minimum inhibitory concentration (MIC) tests against multiple pathogenic strains such as *Shigella dysenteriae*, *Salmonella enterica* serovar Typhimurium, *Escherichia coli*, *Streptococcus pyogenes*, *Staphylococcus epidermidis*, and *Listeria monocytogenes* (Ebadi Nezhad et al., 2020; Nami et al., 2015). To prepare for the antimicrobial activity assay, the *L. acidophilus* VBFD771 strain was cultured in MRS broth. Following incubation, the culture was centrifuged to isolate the cell-free supernatant (CFS). The resulting CFS, in order to mitigate the effects of organic acids, was subsequently prepared in two forms; acidified (aCFS; pH 5.5) and neutral (nCFS; pH 7). In the final step, the CFS was sterilized by filtration and subsequently freeze-dried.

To evaluate the effect of *Lactobacillus* CFS on biofilm formation, an overnight culture of *L. monocytogenes* (5 × 10⁵ CFU/mL) was inoculated into 96-well plates containing the 2 × MIC dose. Following incubation, biofilms were washed with PBS and fixed with 99% methanol. They were then stained with 0.1% crystal violet solution and dissolved in 95% ethanol; quantification was performed by measuring the OD at 570 nm (Sornsene et al., 2021). Additionally, expression of biofilm-associated genes in CFS-treated *L. monocytogenes* (Table 1) was determined by qRT-PCR (Rouhi et al., 2024). Total RNA was extracted (TRIzol) and cDNA synthesized (SensiFAST kit). The specificity of the qRT-PCR was confirmed via melt curve analysis. Relative expression was calculated using the comparative Ct method (2^{-ΔΔCt}) (Echresh et al., 2024; Kim et al., 2020).

2.2.11. Antioxidant effect

The 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging assay was conducted following the method of Jeong et al. (2021), with minor modification. The selected strain was cultured, washed twice with PBS, and resuspended at a known concentration. This LAB sample was mixed with a 0.05 mM DPPH solution in a 1:2 ratio and incubated in the dark for 30 min, with PBS as the control. Absorbance at 517 nm was measured for each sample, and the scavenging activity was calculated accordingly:

$$\text{Scavenging effect}\% = (A_{\text{control}} - A_{\text{sample}}) / A_{\text{control}} \times 100$$

The radical scavenging activity of 2,2'-Azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) diammonium salt (ABTS) was evaluated using a method from Yang et al. (2019), which involved creating an ABTS solution by mixing 7 mM ABTS with potassium persulfate and diluting it in a sodium phosphate buffer to achieve an absorbance of 0.7 at 734 nm. Then, 150 μL of bacterial samples were mixed with 150 μL of the ABTS solution, with distilled water as the control, and incubated at 37 °C for 10 min. The absorbance of these mixtures was measured at 734 nm for calculation of the scavenging activity:

$$\text{Scavenging effect}\% = (A_{\text{control}} - A_{\text{sample}}) / A_{\text{control}} \times 100$$

Table 1
Primers used in this study (Echresh et al., 2024).

Gene	Primer
flaA	Forward CTGGTATGAGTCGCCTTAG
	Reverse CATTTCGGTGTTTGTTTGG
inlB	Forward AAGCAMGATTCATGGGAGAGT
	Reverse TTACCGTTCATCAACATCATAACTT
sigB	Forward GATGATGGATTGAACGTGTGAA
	Reverse CGCTCATCTAAACAGGGAGAAC
agrA	Forward ATGAAGCAAGCGGAAGAAC
	Reverse TACGACCTGTGACAACGATAAA
hly	Forward AACCAGATGTTCTCCCTGTA
	Reverse CACTGTAAGCCATTTTCGTCA
prfA	Forward CGGGAAGCTTGGCTCTATTTC
	Reverse GCTAACAGCTGAGCTATGTGC
plcB	Forward CAGGCTACCACTGTGCATATGAA
	Reverse CCATGTCCTTCYGTGCTTGATAATTG

2.2.12. Cytotoxicity

The assessment of the cytotoxic effect of the CFS from the *L. acidophilus* VBFDP771 strain on the HT-29, HeLa, and MCF-7 human cancer cell lines (Iranian Biological Research Center (IBRC)) was conducted using the MTT technique. First, cells were plated in 96-well microplates at a density of 3×10^5 cells mL⁻¹. Subsequently, various concentrations of CFS (ranging from 0 to 500 mg/mL) were administered, and incubation was carried out for 48 h at 37 °C. Following the incubation period, MTT (4,5-Dimethylthiazol-2-yl)-2,5-Diphenyltetrazolium bromide) was added, and after a further 4 h of incubation, the supernatant was removed. The resulting formazan crystals were dissolved in dimethyl sulfoxide (DMSO), and the absorbance at 570 nm was measured. The percentage of cell viability was then calculated based on the absorbance of the treated and control cells (Liu & Pan, 2010).

2.2.13. Cholesterol assimilation

The cholesterol-lowering activity of the *L. acidophilus* VBFDP771 strain was investigated in MRS liquid broth enriched with cholesterol and bile salts, in accordance with the methodology published by Palaniyandi et al. (2020). To this end, the MRS broth was supplemented with 100 µg/mL cholesterol-polyethylene glycol 600 and 0.3% bile salts. The enriched medium was inoculated with 1% (v/v) of the bacterial culture (10⁶ CFU/mL) and was incubated for 24 h at 37 °C under anaerobic conditions. Following the incubation, the cells were separated by centrifugation (9000 × g for 15 min), and the supernatant containing the non-assimilated cholesterol was collected. Finally, the remaining cholesterol in the supernatant was measured using the o-phthalaldehyde colorimetric method to determine the level of cholesterol assimilation by the strain. In this assay, an uninoculated control sample was also considered the positive control.

2.2.14. Safety evaluation

The safety profile of the *L. acidophilus* VBFDP771 strain was evaluated by assaying for antibiotic susceptibility, hemolytic and DNase activities, and the ability to synthesize biogenic amines.

The antibiotic susceptibility of *L. acidophilus* VBFDP771 to the widely-used antibiotics (chloramphenicol, nitrofurantoin, nalidixic acid, erythromycin, ampicillin, imipenem, and ciprofloxacin) was evaluated using the disc diffusion method on agar plates. Bacteria were incubated overnight in MRS broth at 37 °C, then their concentration was standardized to 10⁸ viable cells. 100 µL were streaked onto MRS agar and antibiotic discs were placed on its surface. After incubation, the size of the inhibition zone around the discs was measured and the results were classified as resistant, semi-sensitive, or sensitive (de Souza et al., 2012).

The hemolysis assay of the *L. acidophilus* VBFDP771 strain was performed by transferring the MRS liquid culture using the streaking method onto sheep blood agar medium (Halder et al., 2017).

The DNase activity assay of the *L. acidophilus* VBFDP771 strain was performed on DNase agar, and the plates were incubated for 24 h at 37 °C. Subsequently, the plates were covered with HCl to precipitate the unhydrolyzed DNA; the clear halo surrounding the colony was recorded as positive activity (Shuhadha et al., 2017).

To evaluate the potential for biogenic amine synthesis such as putrescine, tyramine, and histamine by the *L. acidophilus* VBFDP771 strain, the test was conducted by adding their amino acid precursors (including L-ornithine, L-tyrosine, and L-histidine) at a 1% concentration to MRS agar plates. The formation of a purple halo in the yellow culture medium indicated the decarboxylation activity and the production of biogenic amines by the strain (Boubakri et al., 2022).

2.3. Gaussian process regression (GPR) model

GPR is a flexible, nonparametric, and probabilistic approach to regression that leverages kernel methods to model complex relationships in data. Rather than imposing a fixed functional form-such as

linearity or polynomial structure- GPR treats the underlying mapping from inputs to outputs as a sample from a Gaussian process (Li et al., 2024).

A distinguishing feature of GPR is its capacity to quantify prediction uncertainty, delivering not only point estimates but also credible intervals that reflect confidence in those predictions- an attribute especially critical in risk-sensitive domains like healthcare, environmental modeling, or autonomous systems. By selecting appropriate kernels-such as the widely used Radial Basis Function (RBF), Matérn, or periodic kernels- practitioners can tailor the model to the specific characteristics of their data. GPR excels in settings with limited, sparse, or noisy observations, where its Bayesian foundation enables robust inference even when data is scarce.

The main equation for GPR model is (Rohani et al., 2018):

$$y = f(x) + \varepsilon, \varepsilon \approx N(0, s_{noise}^2) \quad (1)$$

the noise variance plays a crucial role in the model. The consistency requirement enables the derivation of function values from a given set of training data. By assuming a Gaussian prior, the function can be expressed through the mean function $m(x)$ and the covariance function.

2.4. Normalization and K-fold cross validation

K-fold cross-validation is a reliable method for assessing the predictive performance of machine learning models. Over the course of these iterations, every fold serves once as the validation set, ensuring comprehensive coverage of the data. The performance scores from all k runs are then aggregated- typically by averaging- to yield a single, more stable estimate of model accuracy (Rohani et al., 2018).

A key strength of this technique lies in its ability to mitigate overfitting by exposing the model to multiple training-validation configurations. Moreover, it guarantees that each observation in the dataset contributes to both training and testing phases, thereby making efficient use of limited data. While the value of k can vary, 5 and 10 are commonly chosen as they offer a practical balance between computational cost, bias, and variance. Due to its robustness in approximating how a model will perform on unseen data, k-fold cross-validation is a staple in real-world machine learning applications (Rohani et al., 2018). In the present study, 10-fold cross-validation was implemented, as illustrated in Fig. 1.

To evaluate the GPR model, three performance metrics were employed: Mean Absolute Percentage Error (MAPE), the coefficient of determination (R²), and Root Mean Square Error (RMSE) (Rohani et al., 2018).

2.5. Statistical analysis

The analysis of variance (ANOVA) of the data was performed based on a completely randomized factorial design, utilizing Minitab software (version 16). Subsequently, Tukey's Post-hoc test (HSD) was used for significant differences between means at a probability level of $p < 0.05$. All experimental data are presented as mean ± standard deviation (SD) from three independent replicates.

3. Results and discussion

3.1. Probiotic properties of *L. acidophilus* VBFDP771

The ability to withstand acidity and bile salts is crucial when choosing a probiotic strain, as these factors significantly influence the survival of LAB within the digestive system, thereby ensuring their effectiveness and viability as dietary supplements. LAB are typically resilient microorganisms that can endure low pH levels, enabling them to survive harsh conditions like food fermentation and human digestion (Bachtarzi et al., 2024). As illustrated in Fig. 2a, the survival rate of

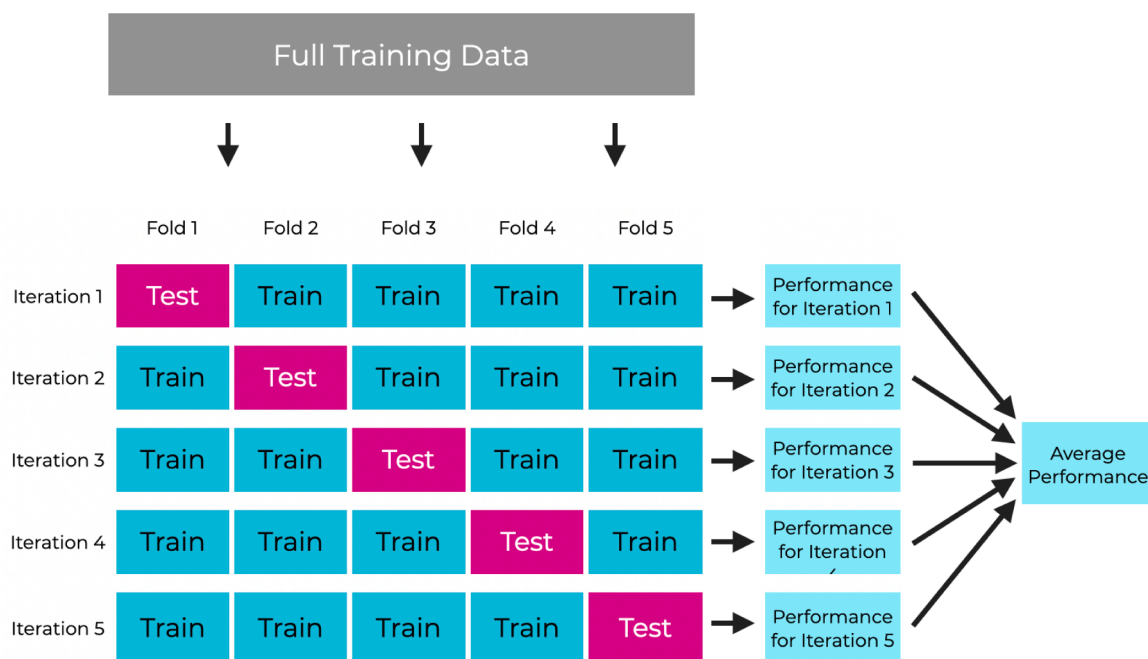


Fig. 1. K-fold cross validation method (Rohani et al., 2018).

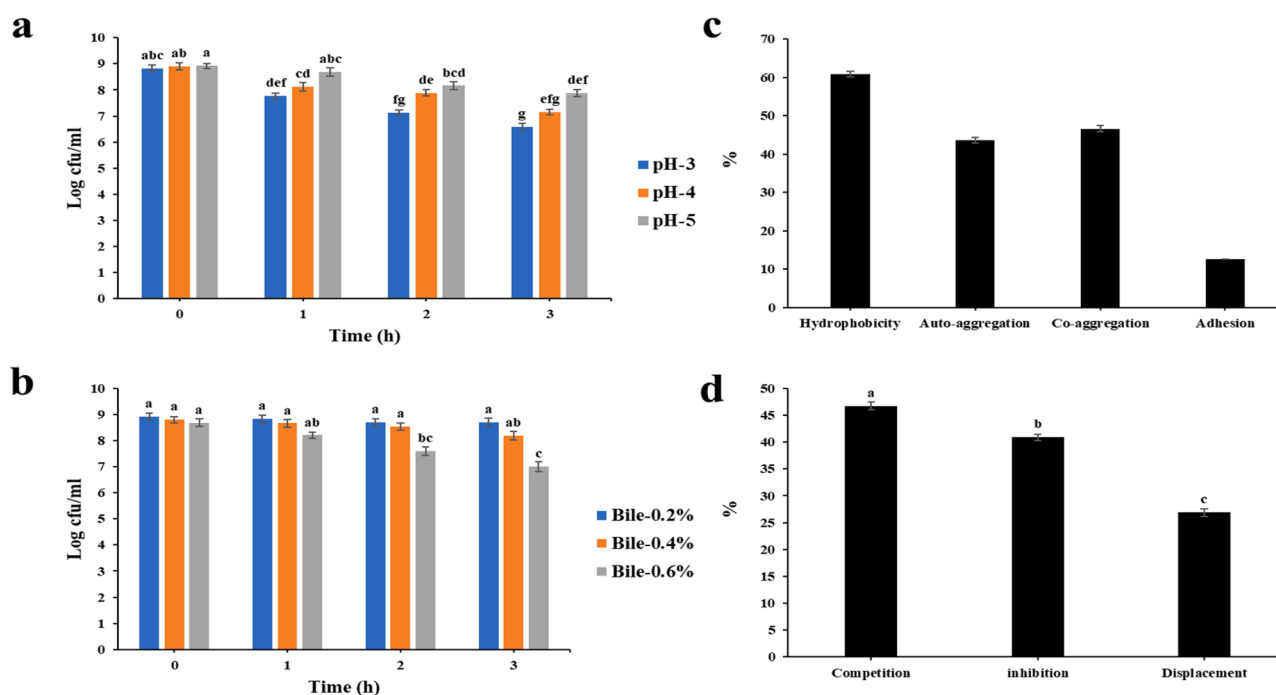


Fig. 2. Survival (log CFU/ml) of *L. acidophilus* VBFDP771 in relation to different acidic pH levels (a) bile concentrations (b). Cell surface (c) and anti-adhesion (d) properties of *L. acidophilus* VBFDP771. Samples that are labeled with different letters indicate significant differences at $p < 0.05$.

L. acidophilus VBFDP771 was examined at various acidic pH levels (3, 4, and 5) over three hours. The data indicated that both the time and decreasing pH negatively impacted bacterial survival, falling from 8.88 to 7.20 log CFU/mL and from 8.41 to 7.57 log CFU/mL, respectively. Furthermore, Fig. 2b explores the growth of *L. acidophilus* VBFDP771 in response to varying concentrations of bile salts. The results demonstrated that this strain remains stable across different bile salt concentrations, although viability declined from 8.78 to 7.87 log CFU/mL as bile salt levels increased. After undergoing simulated gastrointestinal conditions, the viable cell count was recorded at 7.45 log CFU/mL.

Similar results have been documented in other studies (Alizadeh Behbahani, Jooyandeh & Rafiee, 2024a, Alizadeh Behbahani, Jooyandeh & Rafiee, 2024b; Millette et al., 2008). The remarkable resilience of *L. acidophilus* VBFDP771 in acidic environments may stem from the continuous pH gradient maintained by the FOF1-ATPase multicomponent system in Gram-positive bacteria. Additionally, LAB have evolved several regulatory strategies to manage acid stress, including maintaining pH stability through metabolic pathways, altering cell membrane components (such as fatty acids), and protecting vital macromolecules (like DNA and proteins) with chaperones that aid in

repairing damaged proteins and proper folding of newly produced proteins. A key trait for selecting probiotic strains is their capability to detoxify bile salts through enzymes known as bile salt hydrolase, which allows them to thrive in the digestive tract (Saboktakin-Rizi et al., 2021).

Adherence and colonization are crucial traits for the effectiveness of probiotics, especially when it comes to stimulating intestinal enterocytes. A close and prolonged interaction with the intestinal lining likely

necessitates the bacterium's adherence to epithelial cells; without this bond, the probiotic may be unable to impact essential physiological functions (Sanders & Klaenhammer, 2001). The surface properties of *L. acidophilus* cells significantly contribute to their probiotic effects and potential health benefits. Assessing hydrophobicity serves as an initial indicator of the ability of probiotic bacteria to adhere to epithelial cells. This characteristic is vital because it influences the initial interactions

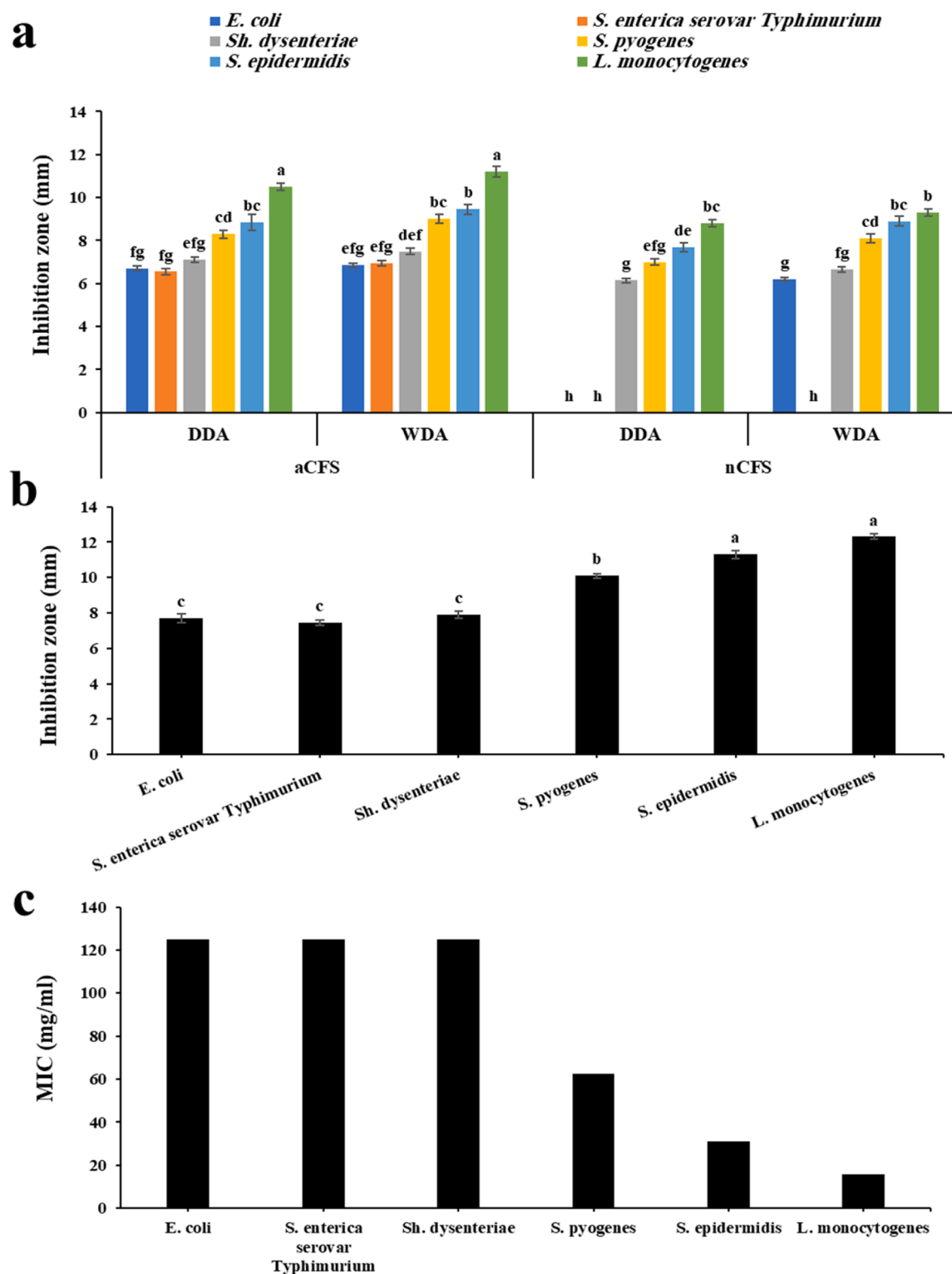


Fig. 3. The antimicrobial effects of cell-free supernatant (CFS) of *L. acidophilus* VBFDP771 based on (a) disk diffusion agar and well diffusion agar, (b) modified double layer, and (c) minimum inhibitory concentration (MIC) methods. Significant differences ($p < 0.05$) are denoted by different superscript letters.

between bacteria and host cells. Bacteria exhibiting higher hydrophobicity tend to adhere more effectively to epithelial cells, enhancing their overall adhesion capabilities. Co-aggregation pertains to close interactions with harmful bacteria, while auto-aggregation is closely connected to the adhesion capabilities of probiotic strains. The ability to adhere is the foremost factor in determining the probiotic efficacy of these strains (Alizadeh Behbahani, Jooyandeh, Taki & Falah, 2024). In the present study, *L. acidophilus* VBFDP771 displayed a notable surface hydrophobicity of 60.85%, along with substantial auto-aggregation at 43.61%, co-aggregation at 46.63%, and an adhesion capacity of 12.63% (Fig. 2c). Another strain of *L. acidophilus* ABC240 exhibited a surface hydrophobicity of 50.49% (Alizadeh Behbahani, Jooyandeh & Rafiee, 2024a, Alizadeh Behbahani, Jooyandeh & Rafiee, 2024b). In a separate study, *L. acidophilus* WFA1 (KU877440), WFA2 (KU877441), and WFA3 (KU877442), sourced from traditional Dahi (yogurt), demonstrated lower adhesion to xylene (54.25%) compared to dichloromethane

(55.25%) and hexadecane (56.65%). WFA1 achieved the highest adherence percentage at 55.48%, followed closely by WFA2 at 55.48% and WFA3 at 51.38%. The levels of cellular auto-aggregation ranged from 21.72% for WFA3 to 30.73% for WFA1 (Farid et al., 2021).

Pathogenic microorganisms require attachment to or invasion of mucosal surfaces to cause disease. In contrast, probiotic bacteria can safeguard the human body against these harmful infections (Alizadeh Behbahani, Jooyandeh, Taki & Falah, 2024). Results demonstrated that *L. acidophilus* VBFDP771 could successfully compete with, inhibit, and prevent the adhesion of *L. monocytogenes* by 46.71%, 40.88%, and 26.86%, respectively (Fig. 2d). Kardooni et al. (2023) reported comparable findings. It is proposed that the anti-adhesion properties of probiotic bacteria stem from various factors, including changes in gut pH, competition with pathogens for nutrients and receptors, the ability of probiotics to enhance the immune response, and the production of antimicrobial substances (Echresh et al., 2024).

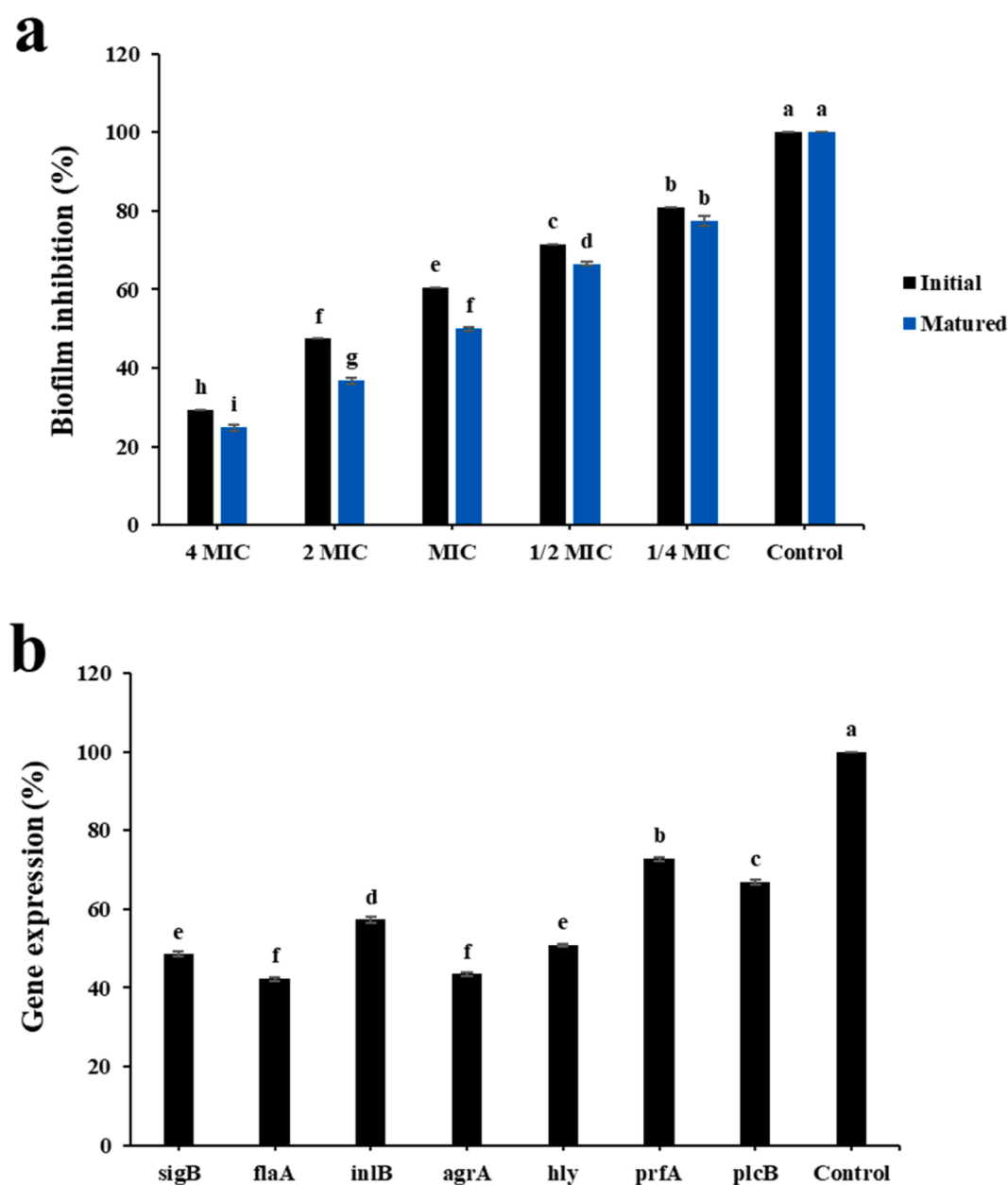


Fig. 4. (a) The antibiofilm activity of the cell-free supernatant (CFS) of *L. acidophilus* VBFDP771 against the biofilm formation capacity of *L. monocytogenes* at both initial and mature biofilm stages. (b) Gene expression analysis in *L. monocytogenes* treated with a sub-inhibitory concentration (1/2 MIC) of the CFS from *L. acidophilus* VBFDP771. Significant differences ($p < 0.05$) are denoted by different superscript letters.

The antibacterial activity of *L. acidophilus* VBFDP771 against various bacterial pathogens is illustrated in Fig. 3. The acidic cell-free supernatant (aCFS) demonstrated greater inhibitory effects on pathogens compared to the neutral cell-free supernatant (nCFS), with inhibition zones (IZs) measuring 8.25 mm for aCFS and 5.73 mm for nCFS ($p < 0.05$). Furthermore, the well diffusion agar method yielded larger IZs (7.51 mm) than those obtained from the disk diffusion agar method (6.47 mm). Among the tested strains, *L. monocytogenes* exhibited the highest sensitivity with an IZ of 9.95 mm, while *S. enterica* serovar Typhimurium showed the least sensitivity with an IZ of 3.37 mm. Additionally, Gram-positive bacteria displayed greater susceptibility to the CFSs than their Gram-negative counterparts. Consistent findings were observed in the minimum inhibitory concentration (MIC) and double layer methods, reinforcing the results from both diffusion techniques.

Treatment with *L. acidophilus* CFS significantly reduced biofilm formation by *L. monocytogenes* in both early and established biofilm stages (Fig. 4a). The expression levels of genes such as *sigB*, *flaA*, *inlB*, *agrA*, *hly*, *prfA*, and *plcB* decreased significantly by 48.50%, 42.32%, 57.30%, 43.46%, 50.85%, 72.73%, and 66.85%, respectively (Fig. 4b). The

reduction in transcription of *plcB*, *hly*, and *prfA* genes suggests that CFS treatment may have lessened the pathogenicity of *L. monocytogenes* by inhibiting hemolysin secretion and modulating phospholipase activity. Comparable findings were reported by *L. acidophilus* ABC240. For instance, Pan et al. (2009) found that *L. acidophilus* and its cell-free supernatant effectively inhibited pathogens like *Salmonella typhimurium*, *Clostridium histolyticum*, *E. coli*, *Bacteroides vulgatus*, and *Clostridium difficile*. Kumar et al. (2016) also demonstrated antimicrobial effects of *L. acidophilus* against multidrug-resistant enteroaggregative *E. coli*. Generally, probiotics replicate the roles of commensal microorganisms through competitive interactions, pathogen antagonism, and the production of antimicrobial agents, including bacteriocin-like compounds such as lactocidin, acidolin, and acidophilin by *L. acidophilus* (Kumar et al., 2016). Research has also shown that *L. acidophilus* NCFM exhibits antagonistic effects against foodborne pathogens such as *S. aureus*, *S. typhimurium*, enteropathogenic *E. coli*, and *Clostridium perfringens*. This study reported a remarkable 80–90% reduction in the growth of these pathogens in broth culture under controlled laboratory conditions. The inhibition is attributed to the production of organic acids, hydrogen peroxide, and possibly other antimicrobial substances (Sanders &

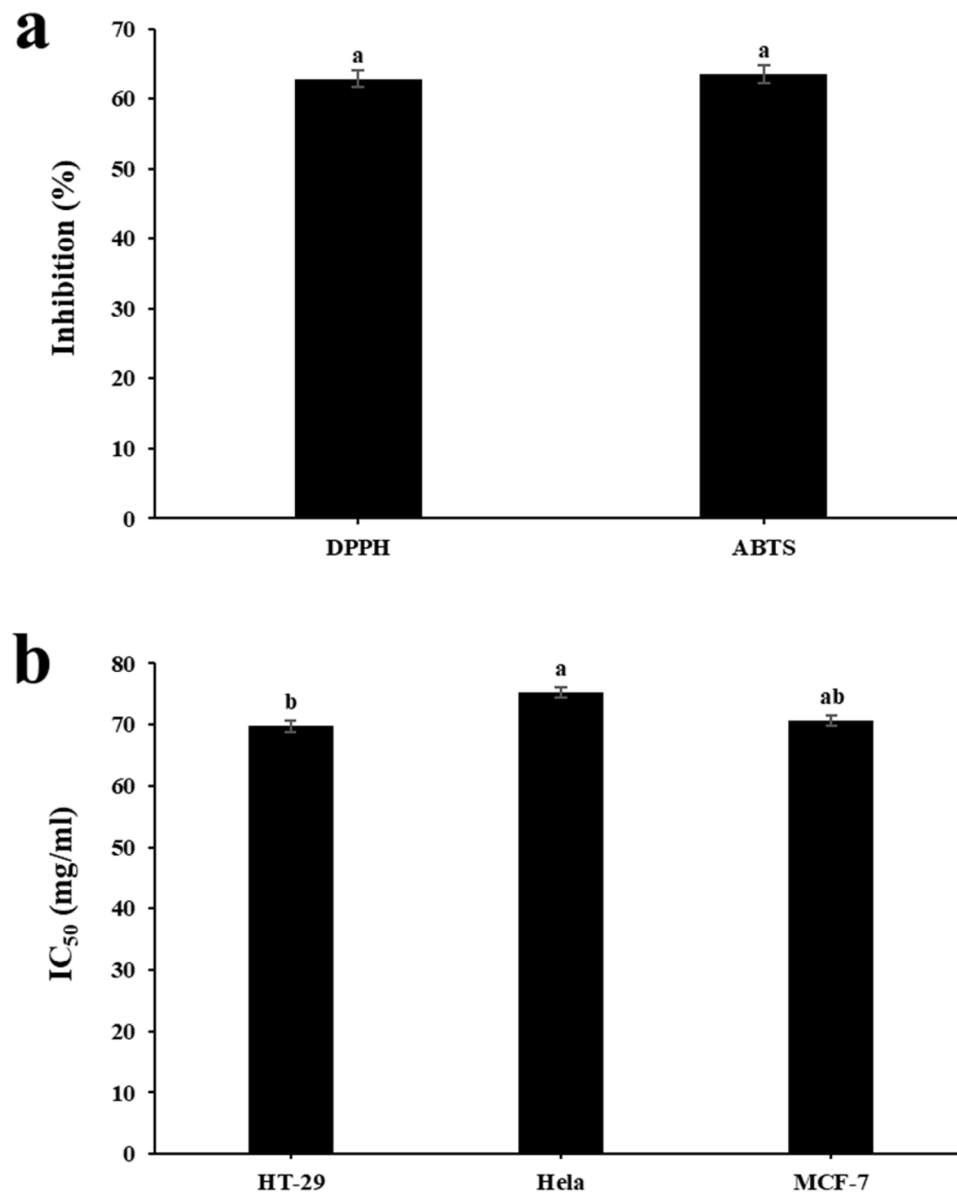


Fig. 5. Antioxidant (a) and cytotoxic (b) effects of *L. acidophilus* VBFDP771. Significant differences ($p < 0.05$) are denoted by different superscript letters.

Klaenhammer, 2001). The antimicrobial efficacy of LAB can be attributed to several factors, including lowered pH levels, competition for nutrients, and the synthesis of bactericidal or bacteriostatic substances like bacteriocins. Specifically, the production of lactic acid reduces pH sufficiently to inhibit some bacterial strains, as the undissociated form of lactic acid decreases the internal pH and disrupts the electrochemical proton gradient in sensitive bacteria, thus exerting a bacteriostatic or bactericidal effect (Pan et al., 2009).

The antioxidant capability is an important trait among *Lactobacillus* strains. *L. acidophilus* VBFDP771 exhibited notable DPPH and ABTS radical scavenging abilities, achieving rates of 62.85% and 63.55%, respectively (Fig. 5a). In a similar vein, Alizadeh Behbahani, Jooyandeh and Rafiee (2024b) found that *L. acidophilus* ABC240 demonstrated impressive antioxidant activity, with its DPPH and ABTS radical scavenging rates measured at 44.69% and 49.80%, respectively. Recent research has identified several *Lactobacillus* strains that show significant antioxidant properties (Alizadeh Behbahani, Noshad, Namazi & Vasiee, 2024; Kim et al., 2022). Probiotics may influence the oxidation state in hosts by binding metal ions and bolstering antioxidant defenses, which can lead to further health advantages (Torshabi, Bardouni & Hashemi, 2023). Additionally, this strain exhibited notable cytotoxicity against cancer cell lines (Fig. 5b), with IC₅₀ values of 69.75, 75.25, and 70.70 mg/mL for HT-29, Hela, and MCF-7 cells, respectively. The anti-cancer effects of two native Iranian strains, *L. acidophilus* and *Lactobacillus paracasei*, were explored both individually and in combination targeting human cervical cancer cell lines, as reported by Samiei Mosleh et al. (2023). The increased expression of apoptotic genes indicated a potential for promoting apoptosis in cancer cells due to the toxicity of *Lactobacillus* (Samiei Mosleh et al., 2023). Like animals and humans, bacteria possess a range of antioxidant enzymes; superoxide dismutase plays a vital role in converting superoxide into water and hydrogen peroxide. Moreover, the antioxidant effects of these bacteria may be associated with their metabolites, such as folate, butyrate, and glutathione (Alizadeh Behbahani, Jooyandeh, Taki & Falah, 2024).

Probiotic bacteria have been shown to effectively reduce cholesterol levels. *L. acidophilus* VBFDP771 demonstrated a cholesterol assimilation rate of 58.80%. Another study reported a rate of 43.60% for *L. acidophilus* ABC240 (Alizadeh Behbahani, Jooyandeh & Rafiee, 2024b). Furthermore, *L. acidophilus* CL1285 displayed activities of both bile salt hydrolase (BSH), feruloyl esterase (FAE), and in vitro cholesterol assimilation (Frappier et al., 2022).

The strain exhibited the strongest resistance to ampicillin, with an

inhibition zone measuring 15.50 mm, while it displayed the weakest resistance to nitrofurantoin, which had an inhibition zone of 26.00 mm (Fig. 6). It was found to be moderately sensitive to ampicillin, but fully susceptible to nitrofurantoin, chloramphenicol, erythromycin, imipenem, nalidixic acid, and ciprofloxacin. Furthermore, *L. acidophilus* VBFDP771 showed no signs of hemolytic activity, DNase production, or the formation of biogenic amines. These results align with the study conducted by Alizadeh Behbahani, Jooyandeh and Rafiee (2024b).

3.2. Gaussian process regression (GPR) model and normalization and K-fold cross validation

In this study, the GPR model was selected to predict acidity and bile salt levels. To enhance the reliability of the results, the K-Fold cross-validation method was employed. Table 2 demonstrates that K-Fold cross-validation can predict both parameters with low Mean Absolute Percentage Error (MAPE) and standard deviation. Specifically, the MAPE for acidity was 1.05 ± 0.35 , and for bile salt, it was 1.72 ± 0.57 , indicating that the GPR model provided acceptable outputs. To identify the best run of the GPR model, the gname command was utilized.

Table 3 presents the results of the best run (No. 80), showing a MAPE of 0.74% for acidity and 1.33% for bile salt prediction. Additionally, the R² values were 0.97 and 0.94 for acidity and bile salt, respectively. Fig. 7 illustrates the distribution of actual and predicted data by the GPR model, revealing that the data points were closely aligned with the regression line, indicating high prediction accuracy.

Furthermore, Table 4 details various statistical indexes, such as average, variance, standard deviation, minimum, maximum, kurtosis, and skewness for both actual and predicted data across all phases. The close alignment of these indexes underscores the model's robustness and accuracy in predicting acidity and bile salt levels. This comprehensive analysis highlights the effectiveness of the GPR model and the reliability of the predictions obtained through K-Fold cross-validation.

4. Conclusion

L. acidophilus strain VBFDP771 emerges as a highly promising probiotic, exhibiting remarkable resilience in acidic conditions and the ability to thrive in gastrointestinal environments. Its potent anti-adhesion effects against *L. monocytogenes* and broad-spectrum antimicrobial activity highlight its potential to prevent pathogenic infections and support gut health. Additionally, this strain displays antioxidant

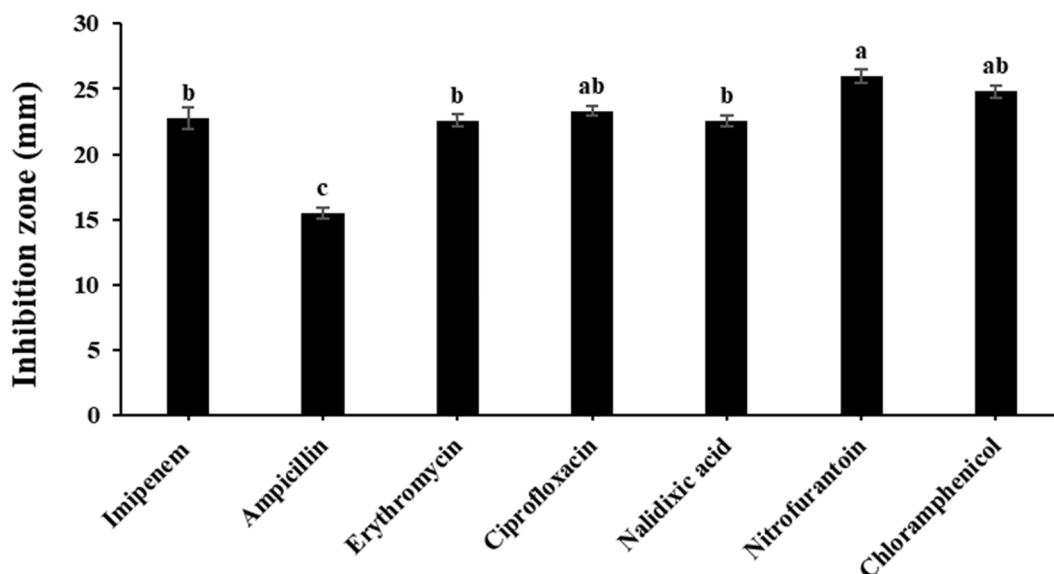


Fig. 6. Antibiotic susceptibility profile of *L. acidophilus* VBFDP771. Significant differences ($p < 0.05$) are denoted by different superscript letters.

Table 2
The results of GPR model with K-Fold cross validation method.

Predicted factors	Train			Test			Total		
	RMSE ($\bar{x} \pm SD$)	MAPE ($\bar{x} \pm SD$)	R ² ($\bar{x} \pm SD$)	RMSE ($\bar{x} \pm SD$)	MAPE ($\bar{x} \pm SD$)	R ² ($\bar{x} \pm SD$)	RMSE ($\bar{x} \pm SD$)	MAPE ($\bar{x} \pm SD$)	R ² ($\bar{x} \pm SD$)
Acidity	0.02 ± 0.03	0.20 ± 0.09	0.97 ± 0.04	0.19 ± 0.06	1.90 ± 0.58	0.97 ± 0.05	0.14 ± 0.05	1.05 ± 0.35	0.97 ± 0.03
Bile salt	0.03 ± 0.06	1.30 ± 0.06	0.99 ± 0.01	0.14 ± 0.11	1.40 ± 1.12	0.90 ± 0.12	0.10 ± 0.08	1.72 ± 0.57	0.94 ± 0.10

Table 3
The results of the GPR model (best run of K-fold cross validation; run No: 80).

Predicted factors	Train			Test			Total		
	RMSE	MAPE	R ²	RMSE	MAPE	R ²	RMSE	MAPE	R ²
Acidity	0.01	0.03	0.97	0.15	1.44	0.97	0.11	0.74	0.97
Bile salt	0.01	0.02	0.98	0.25	2.64	0.95	0.17	1.33	0.94

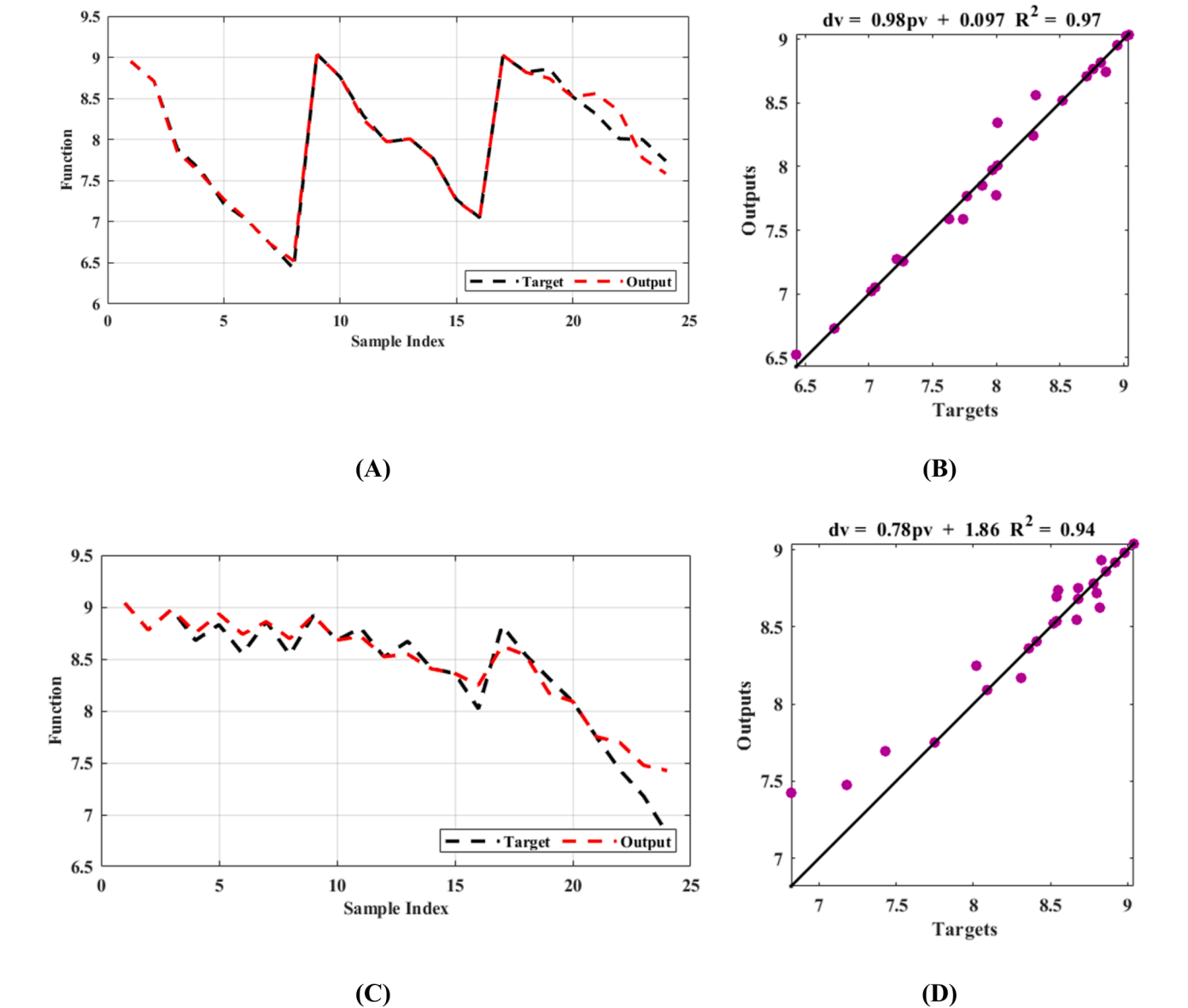


Fig. 7. The results of GPR model; Selection the best run of K-fold cross validation method for acidity (A and B) and bile salt (C and D) prediction.

properties, cholesterol-lowering benefits, and a solid safety profile, making it an excellent candidate for health supplements and functional foods. In addition, the GPR model was used to predict acidity and bile salt levels, with K-Fold cross-validation enhancing reliability. The MAPE

Table 4
Actual and predicted data values of GPR model.

Parameters	Phases	Data	Average	Variance	Standard deviation	Minimum	Maximum	Kurtosis	Skewness	Sum
Acidity	Train	Actual	8.12	0.68	0.83	6.73	9.04	1.79	−0.48	97.42
		Predicted	8.11	0.68	0.83	6.71	9.03	1.80	−0.48	97.39
	Test	Actual	7.88	0.50	0.71	6.43	8.95	2.82	−0.36	94.60
		Predicted	7.89	0.50	0.71	6.52	8.95	2.35	−0.23	94.70
	Total	Actual	8.00	0.58	0.76	6.43	9.04	2.14	−0.35	192.02
		Predicted	8.00	0.58	0.76	6.52	9.03	1.95	−0.31	192.12
Bile salt	Train	Actual	8.58	0.15	0.38	7.75	9.04	2.82	−0.77	102.91
		Predicted	8.58	0.15	0.38	7.62	9.03	2.78	−0.77	102.54
	Test	Actual	8.22	0.49	0.70	6.82	8.83	2.42	−0.96	98.65
		Predicted	8.34	0.28	0.53	7.42	8.93	2.02	−0.74	100.02
	Total	Actual	8.40	0.34	0.58	6.82	9.04	3.97	−1.34	201.58
		Predicted	8.46	0.22	0.47	7.42	9.04	2.80	−0.93	202.95

for acidity was 1.05 ± 0.35 , and for bile salt, it was 1.72 ± 0.57 , indicating acceptable outputs. The best run (No. 80) showed a MAPE of 0.74% for acidity and 1.33% for bile salt, with R^2 values of 0.97 and 0.94, respectively.

Consent for publication

All authors approved the manuscript for publication.

Availability of data and material

All data relevant to the study are included in the article.

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Ethical approval

This study was approved by the Agricultural Sciences and Natural Resources University of Khuzestan. All methods were carried out in accordance with relevant guidelines and regulations.

Clinical trial number

Not applicable.

Consent to participate declaration

Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process

In preparing this work, the authors utilized "Microsoft Copilot" to assist with paraphrasing and grammar checks. After employing this tool/service, the authors carefully reviewed and edited the content as necessary and take full responsibility for the content of the published article.

Ethical statement

This study did not involve human participants or experimental animals. All experiments were conducted in vitro using bacterial strains and established cell lines. Therefore, ethical approval from an institutional review board or ethics committee was not required. All procedures were performed in accordance with relevant institutional and international guidelines for laboratory research.

CRedit authorship contribution statement

Behrooz Alizadeh Behbahani: Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization. **Hossein Jooyandeh:** Writing – original draft, Validation. **Morteza Taki:** Validation, Software. **Alireza Vasiee:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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